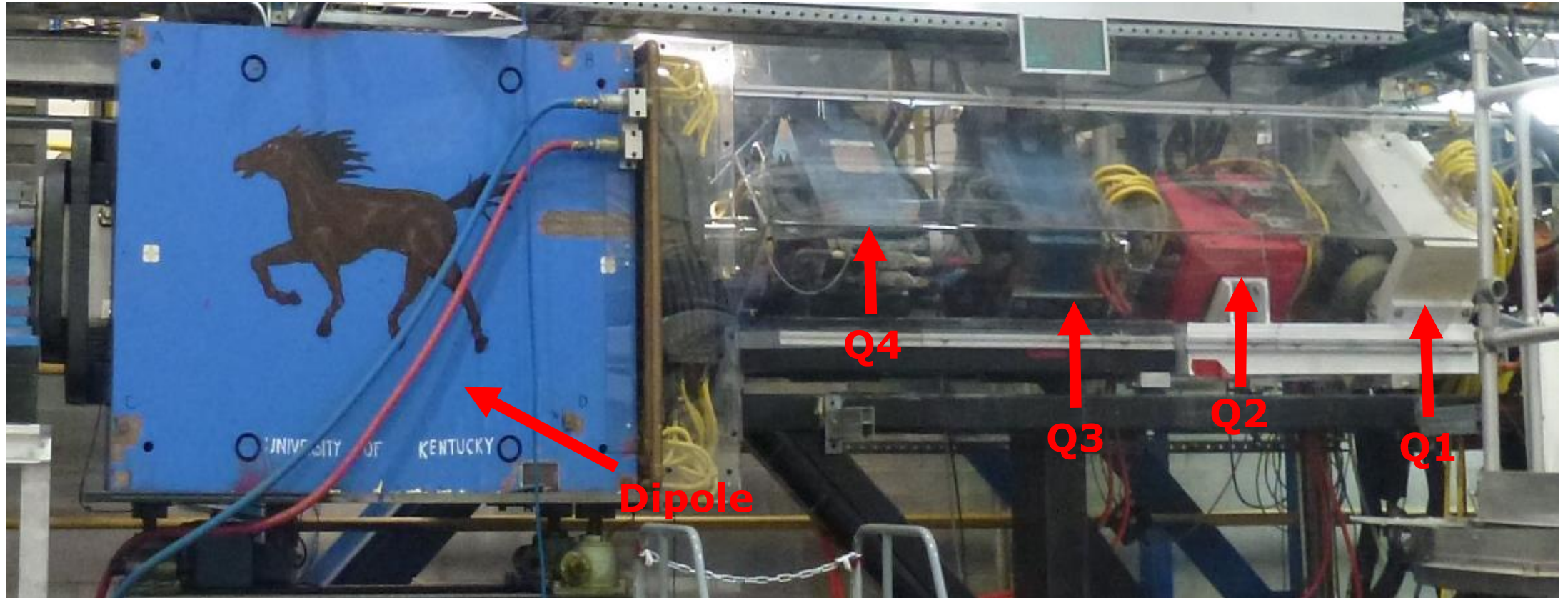


***TOSCA FIELD MAP
FOR
MOLLER MAGNETS***

Sasha Glamazdin

Moller Polarimeter Magnets



Quads Information from Los-Alamos

Møller notation	Q1 /Q2	Q2 /Q3	Q3 /Q4	-	Dipole
Name	<i>PATSY</i>	<i>TESSA</i>	<i>FELICIA</i>	<i>JACKIE</i>	<i>LILLY</i>
Relevant information					
Element	quad	quad	quad	quad	dipole
Bore, cm	10.16	10.16	10.16	10.16	
Effective length, cm	45.72	35.66	35.66	45.72	164.27
Maximum current, A	300	280	280	300	550
Pole tip field at 100A, kGs	2.029	2.371	2.420	≈2.03	2.727
Pole tip field at 300A, kGs	5.801	6.029	6.135	5.809	
Integrated GL at 100A, Gs	18308	17112	17524	≈18300	
Other technical information					
Manufacturer	MagnaTek	Indust.Coil	Indust.Coil		
Part number	LA-446205	LA-446201	LA-446200		
Yoke type	Solid	Solid	Solid		
Yoke material	1006	1006	1006		
Weight, kg	450	500	500		
Color	Red	Blue	Blue	Blue	
Number of coils	4	4	4	4	4
Num. of turns per coil	42	50	50	42	32
Resistance @20C°, ohm	0.1133	0.0767	0.0767		
Shunts and sec. coils	none	none	none	none	none
Voltage drop at max cur	10.6VDC	8.5VDC	8.5VDC		
Total water flow	1.5gpm	1.0gpm	1.0gpm		
Max. oper. pressure	35psi	35psi	35psi		
Cooling medium	LCW	LCW	LCW		

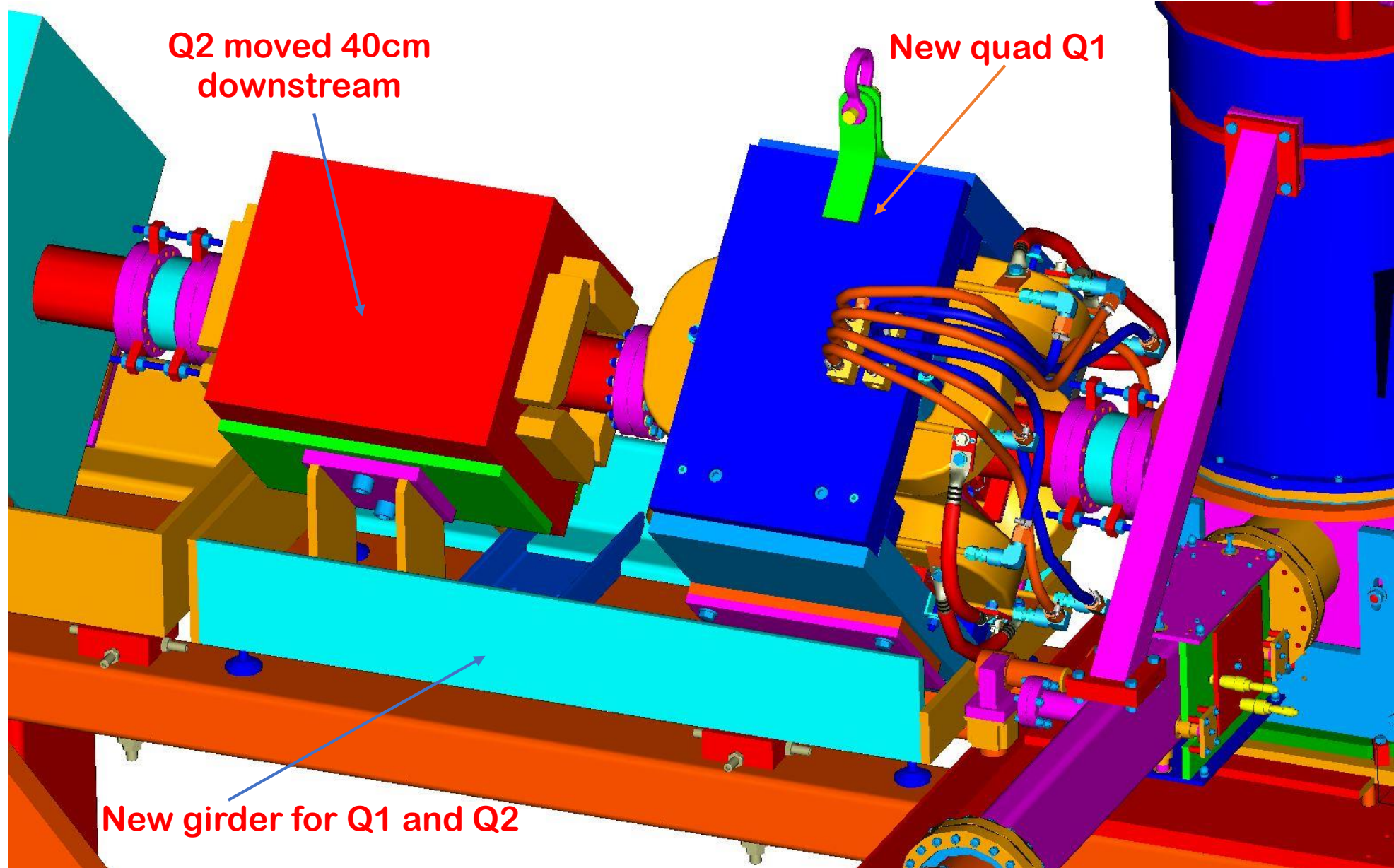
More info on: <http://hallaweb.jlab.org/equipment/moller/spectrometer.html>

Quads Information from Los-Alamos

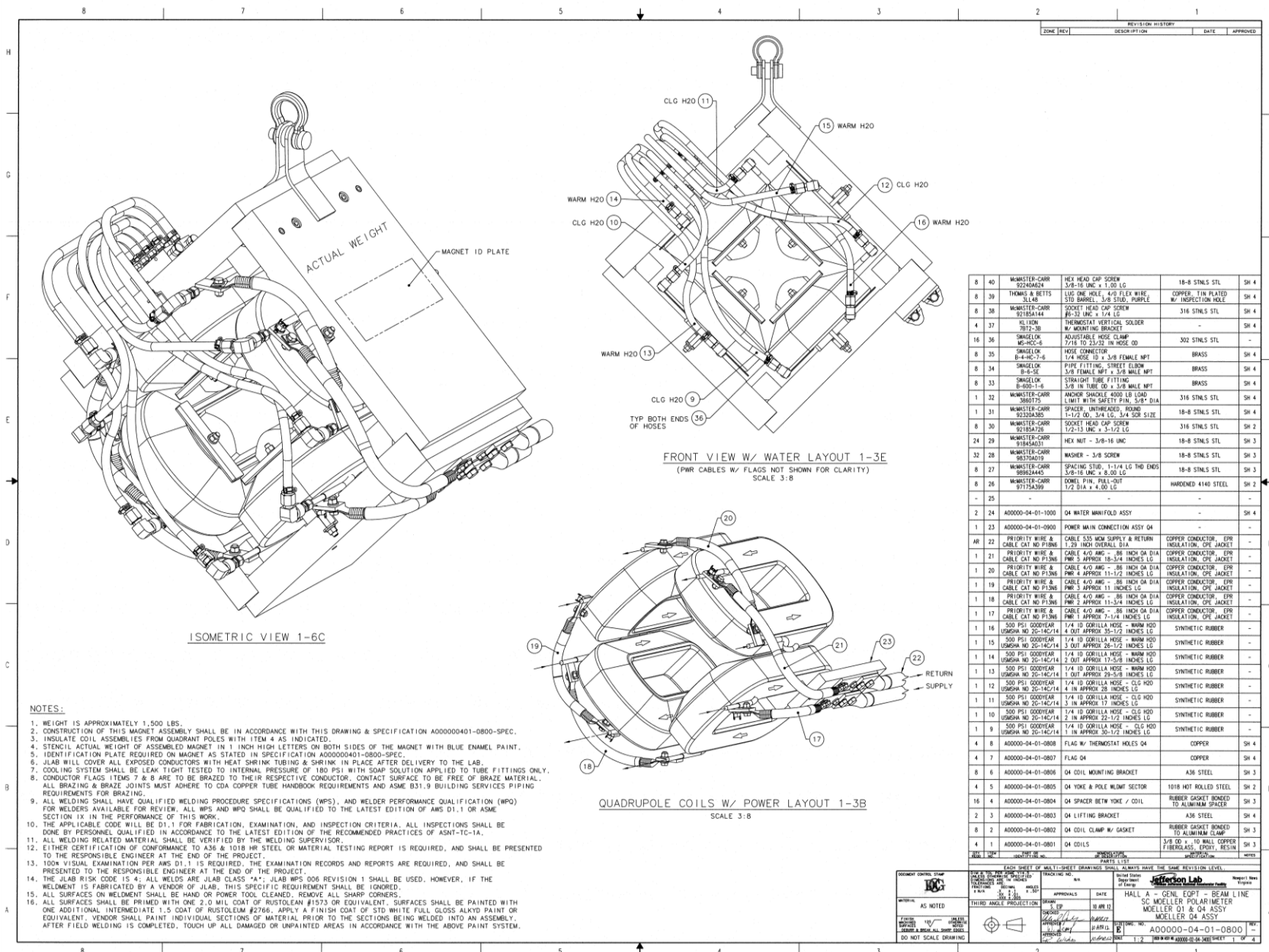
Table 1: Hall A Moller Quadrupole Parameters

Quad 1 Radius= 4.96cm Length= 44.77cm /Q2						
Current	B _{pole}	fG.dl (Gauss)				
(A)	(Gauss)	Quadrupole	Sextupole	Octupole	Decapole	Dodecapole
0.00	32.00	290.00	0.32	0.17	0.03	1.51
49.94	1014.50	9143.00	10.97	1.83	1.83	16.46
100.24	1997.00	18070.00	28.91	3.61	3.61	32.53
150.14	2976.00	26910.00	40.36	5.38	5.38	48.44
200.09	3954.00	35710.00	53.57	14.28	7.14	74.99
249.93	4918.00	44350.00	70.96	22.17	8.87	115.31
299.15	5801.50	52040.00	98.88	26.02	10.41	161.32
Quad 2 Radius= 5.00cm Length= 36.74cm /Q3						
Current	B _{pole}	fG.dl (Gauss)				
(A)	(Gauss)	Quadrupole	Sextupole	Octupole	Decapole	Dodecapole
0.00	31.00	228.00	1.35	0.09	0.16	1.48
50.56	1187.00	8789.00	25.49	0.88	1.76	23.73
100.26	2329.50	17140.00	63.42	5.14	1.71	47.99
150.08	3450.00	25340.00	101.36	5.07	2.53	68.42
200.04	4530.00	33250.00	139.65	6.65	6.65	93.10
250.20	5418.50	39750.00	155.03	7.95	7.95	123.22
299.04	6029.50	43980.00	153.93	8.80	8.80	145.13
Quad 3 Radius= 5.00cm Length= 36.50cm /Q4						
Current	B _{pole}	fG.dl (Gauss)				
(A)	(Gauss)	Quadrupole	Sextupole	Octupole	Decapole	Dodecapole
0.00	30.00	229.00	1.92	0.25	0.25	1.56
50.11	1208.00	8783.00	21.96	0.88	0.88	22.84
100.28	2379.00	17290.00	43.23	1.73	1.73	43.23
150.07	3528.50	25570.00	63.92	2.56	2.56	66.48
199.95	4634.00	33540.00	83.85	6.71	3.35	97.27
249.99	5533.50	40030.00	100.08	8.01	8.01	128.10
299.03	6135.00	44240.00	110.60	8.85	4.42	154.84

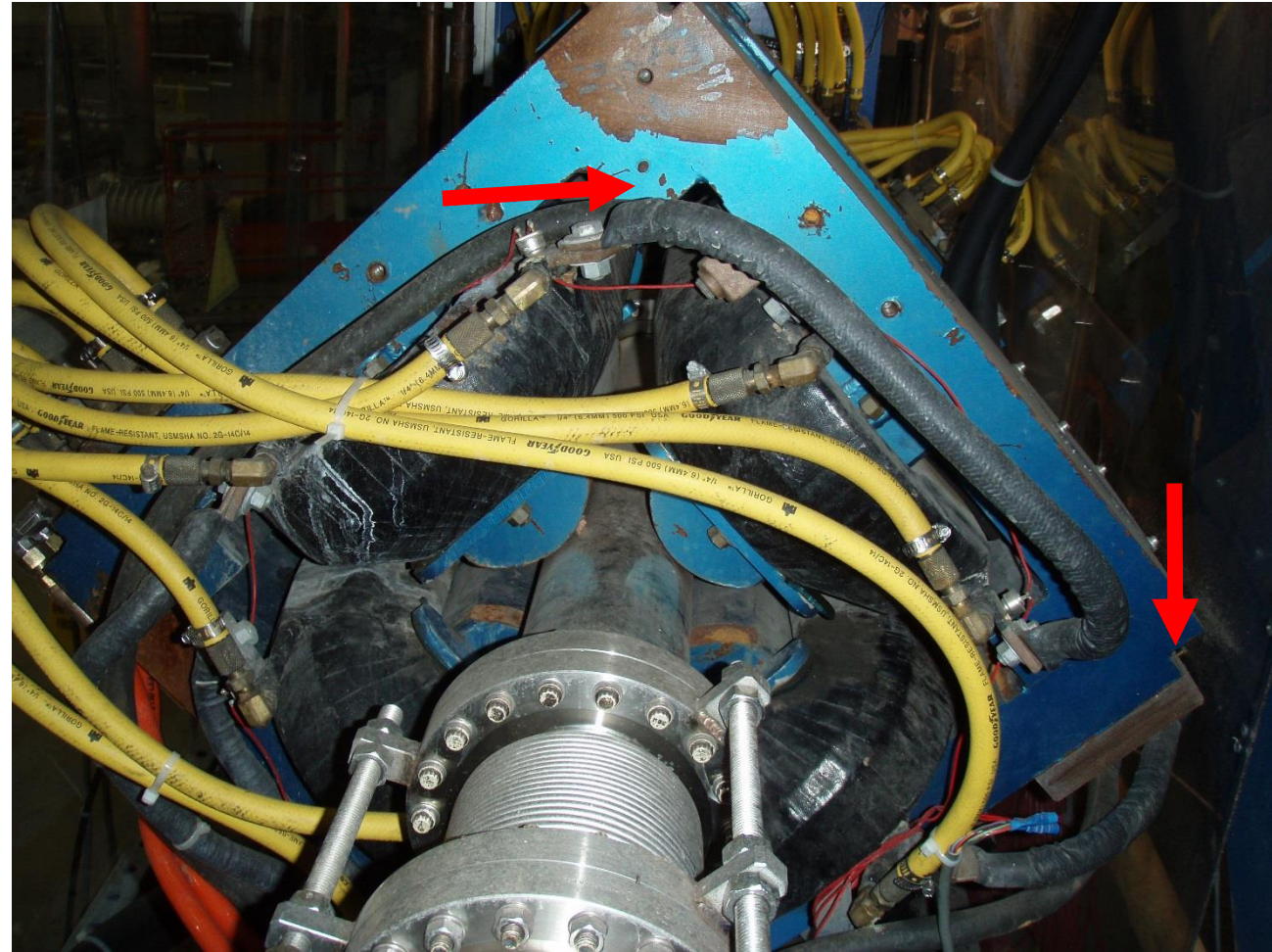
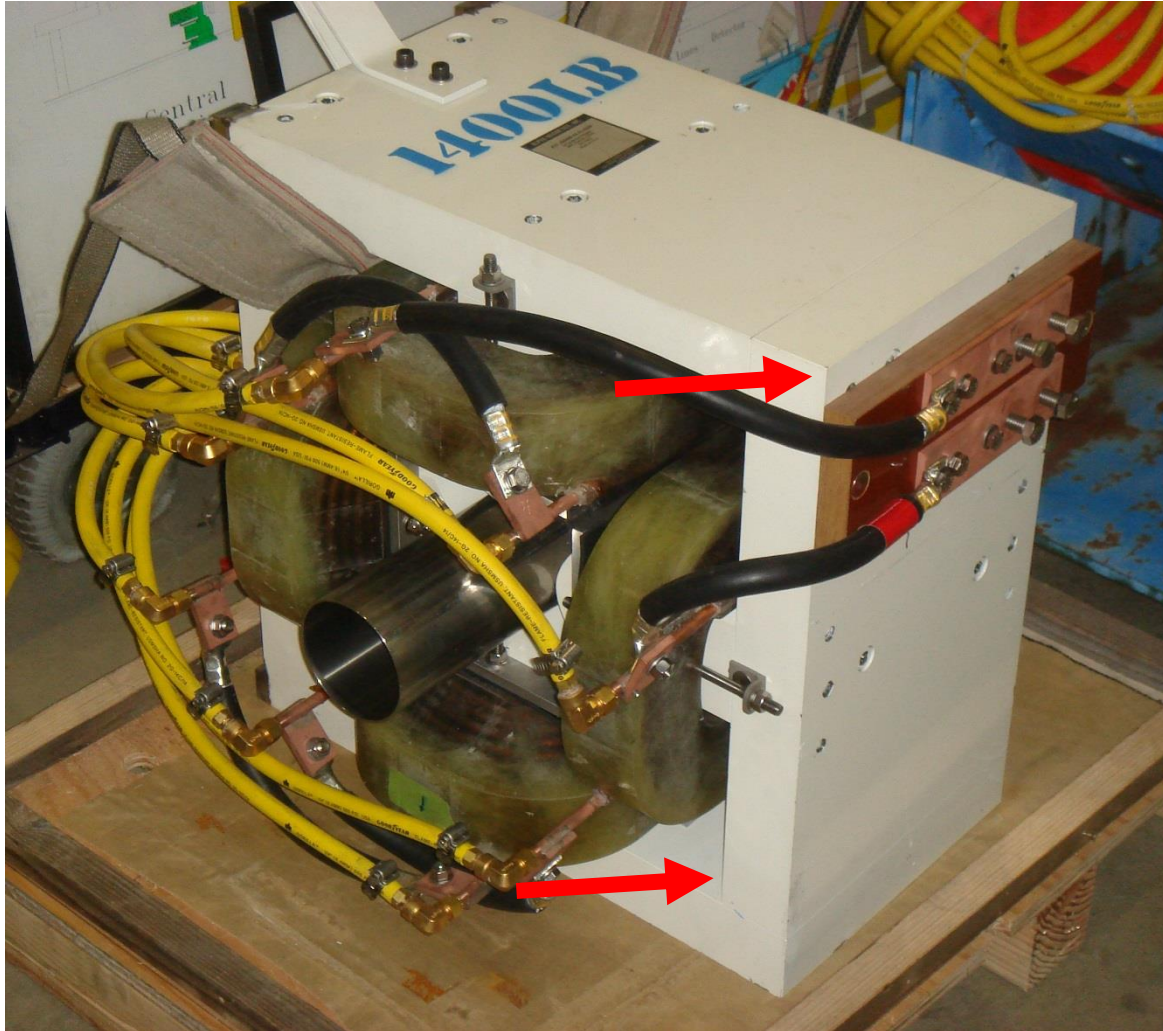
Quads Upgrade



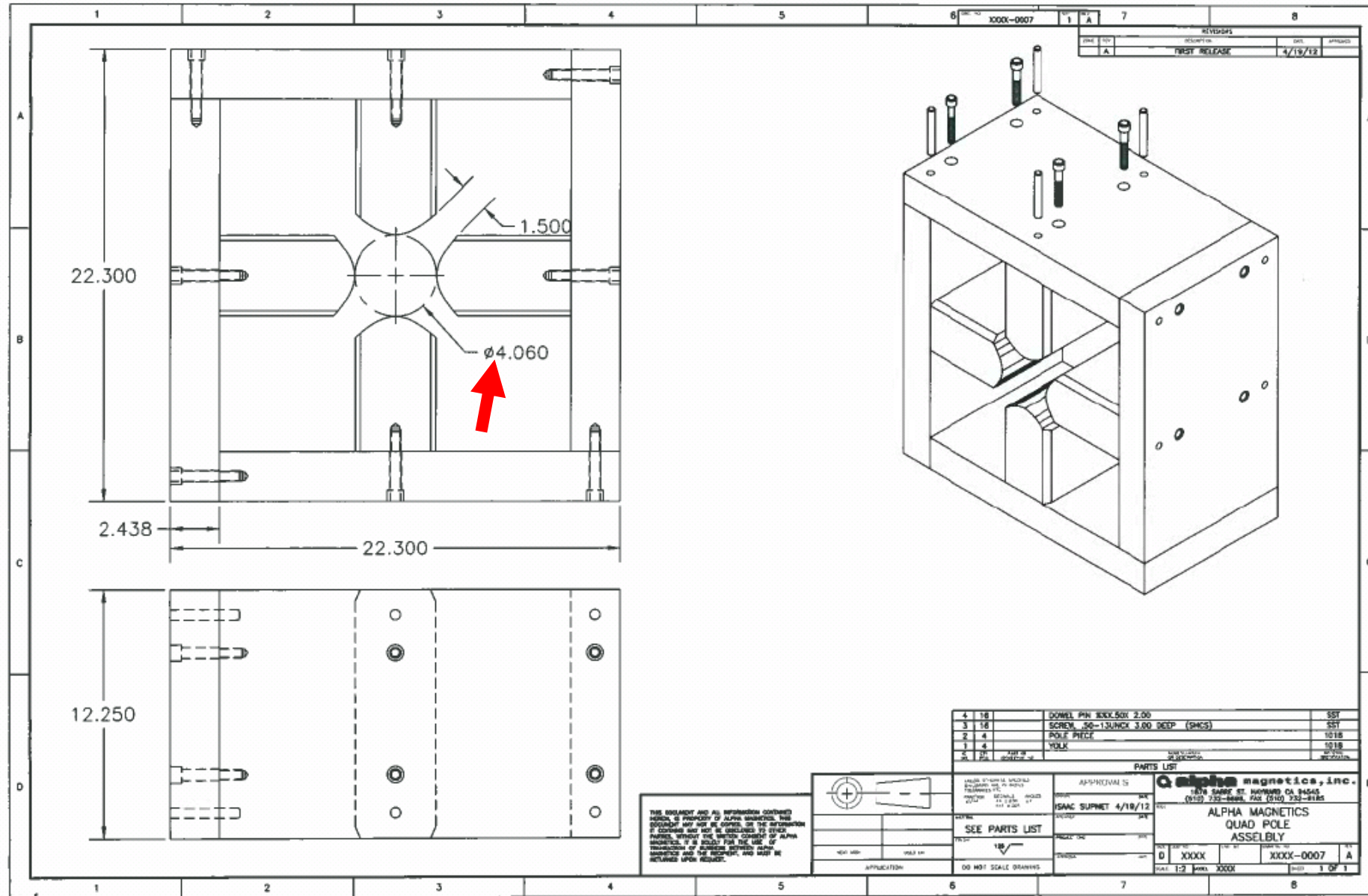
New Quad Q1 Drawing



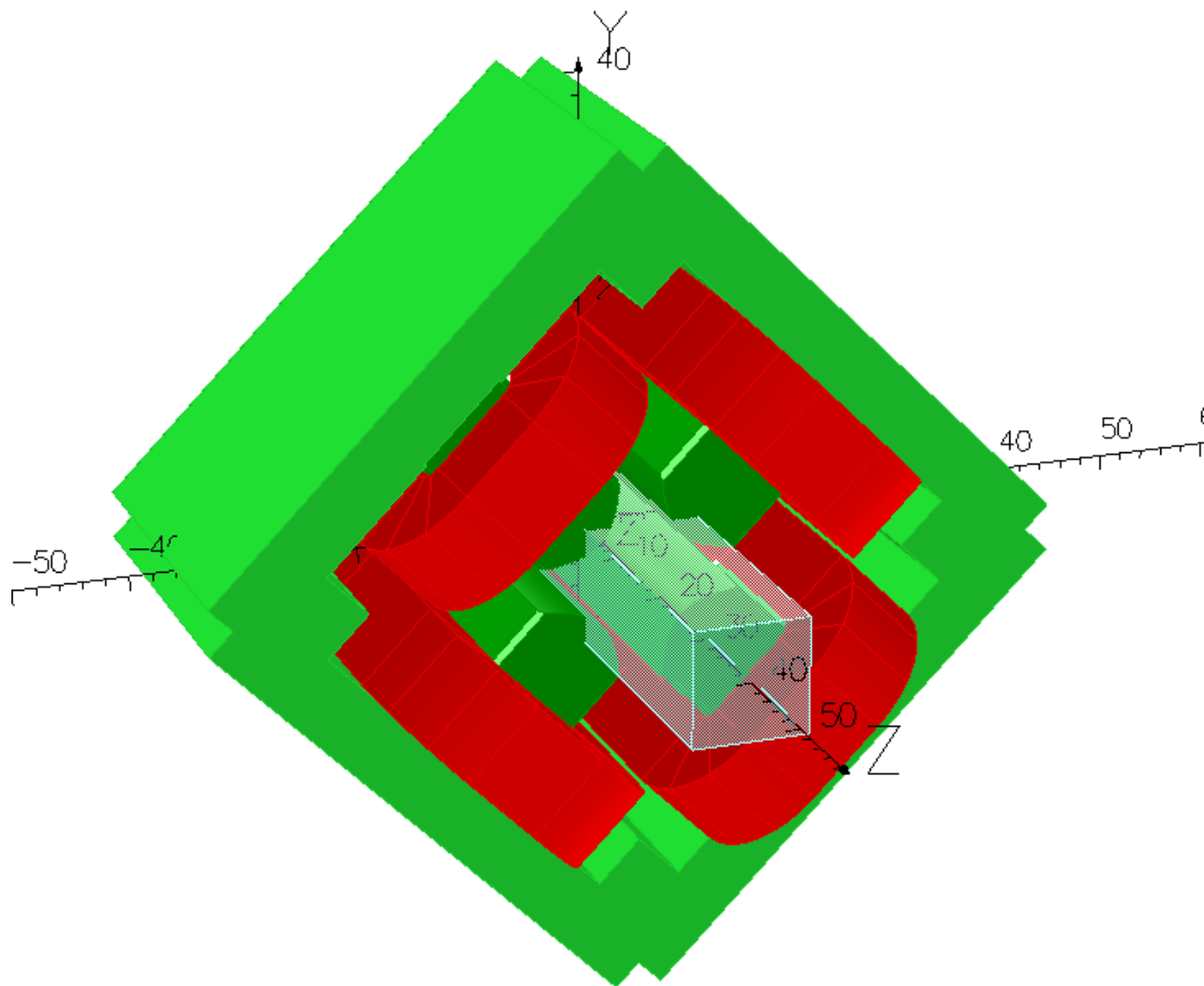
New Quad Q1 and old Q3



New Quad Q1: Vendor Updated



Quads Q1/Q3/Q4



Q1 (MQO1H01) New Moller quad

From Ken Bagget (2013)

$I(A)$	$R(cm)$	$BL(kG*cm)$	$B(kG)$	$L_{eff}(cm)$
299.962	4.035	188.369	5.149	36.58

Maximal Field at pole:

$$B = 5.149 / 4.035 * 5.08 = 6.48 kG$$

TOSCA:

$$242 A/cm^2 \rightarrow 15016 A / 50 turns = 300.32 A$$

Max Field (on pole tip $R=5.08cm$: 6.469kG

TOSCA grid file: q1_6.47kG.table

X: -5cm ÷ +5cm step 0.25cm

Y: -5cm ÷ +5cm step 0.25cm

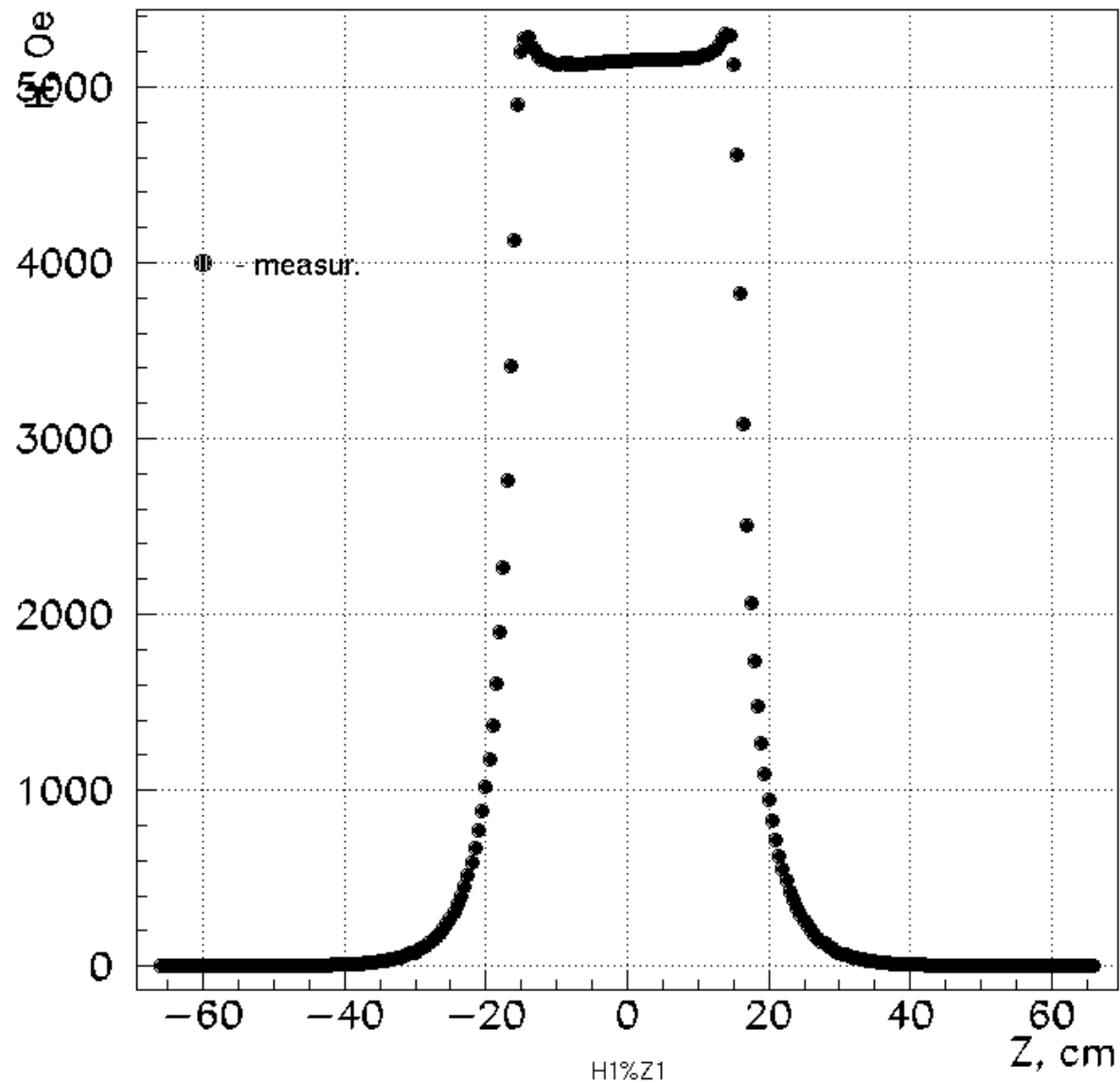
Z: -40cm ÷ +40cm step 0.5cm

File size 38MB, Location: /u/scratch/glamazdi

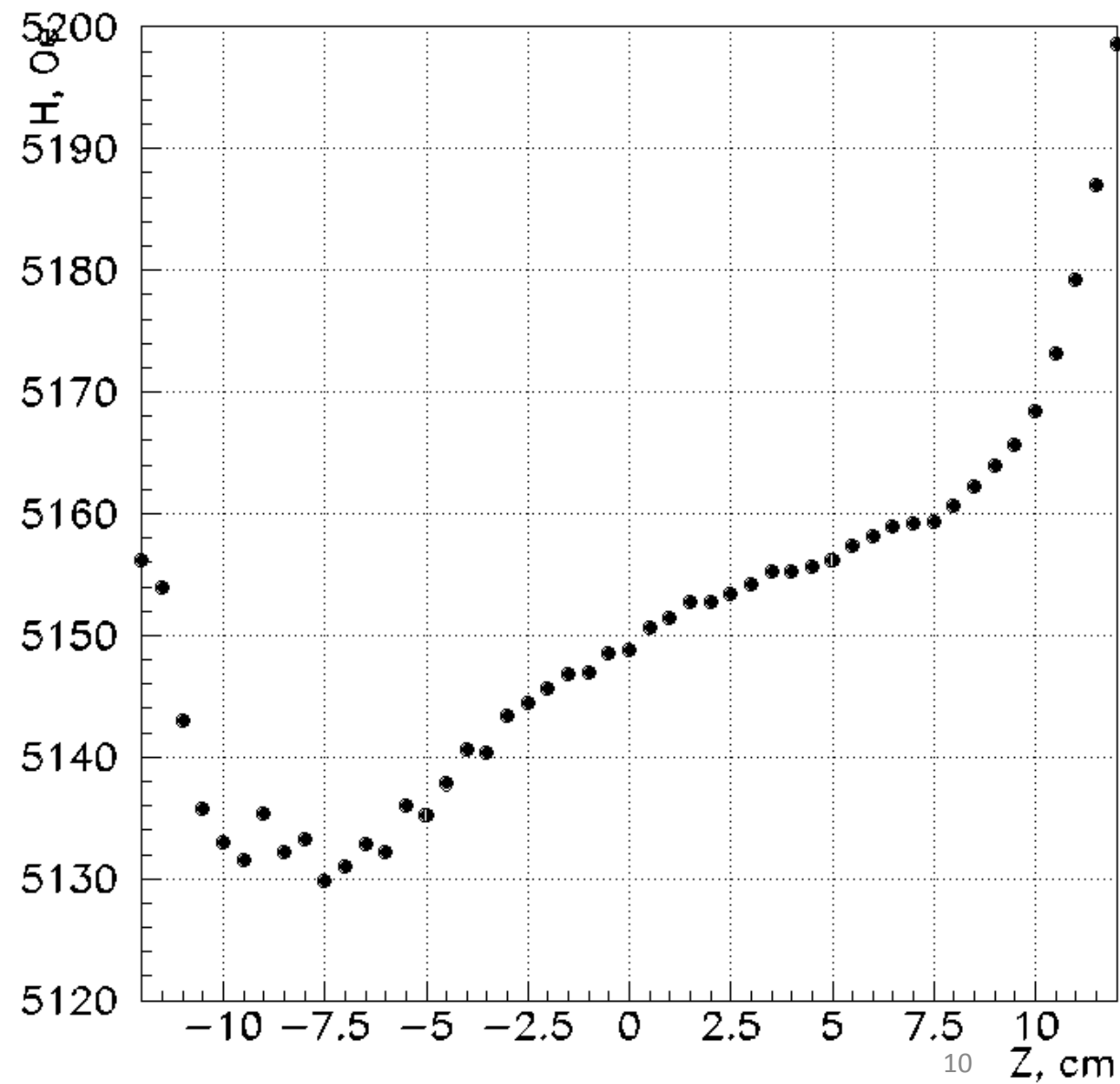
**Q1 body is about identical to Q3 and Q4.
Field map can be used for all three quads.**

New Quad Q1: measurements

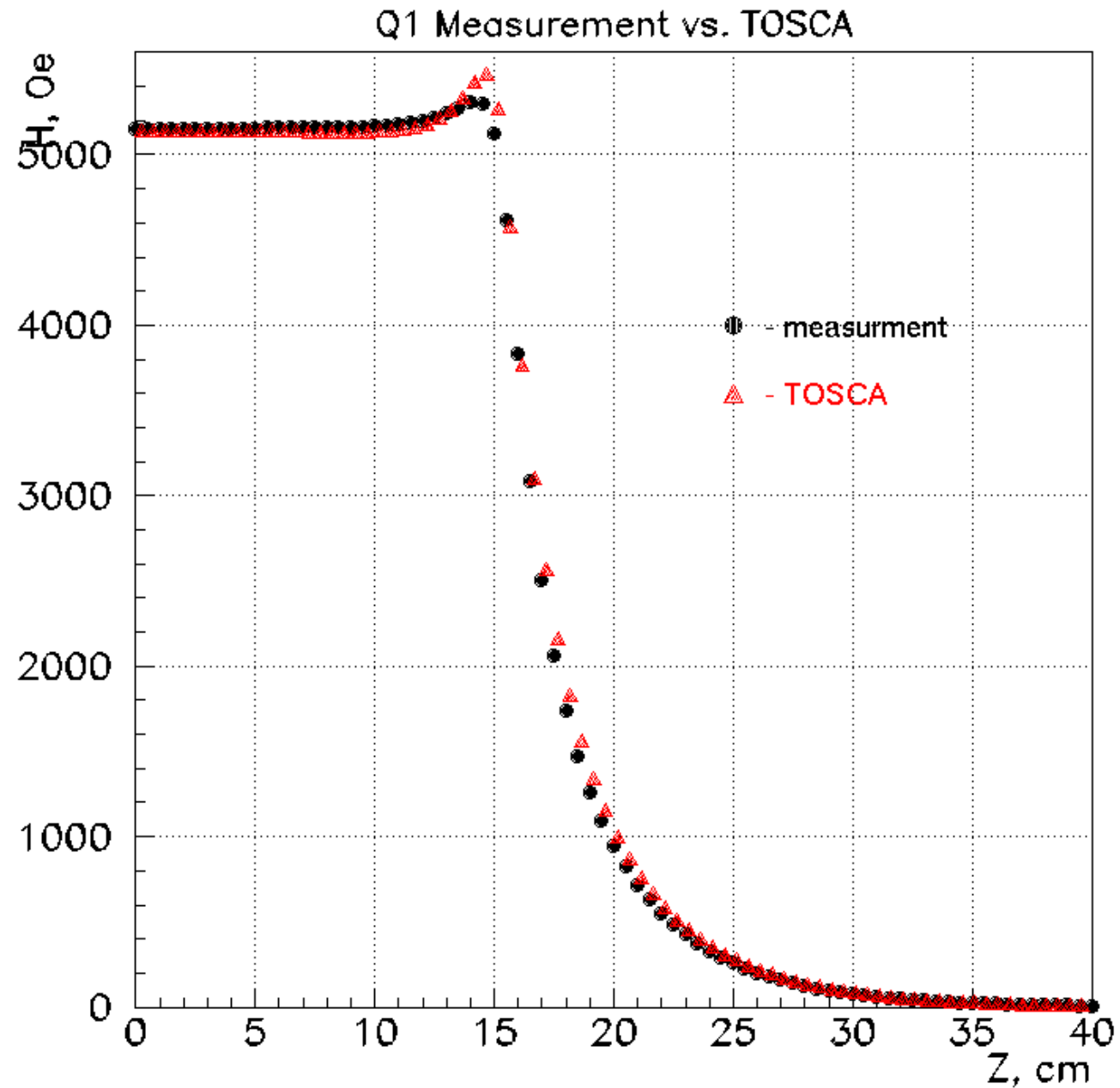
Q1 Measurement vs. TOSCA



Q1 Measurement vs. TOSCA



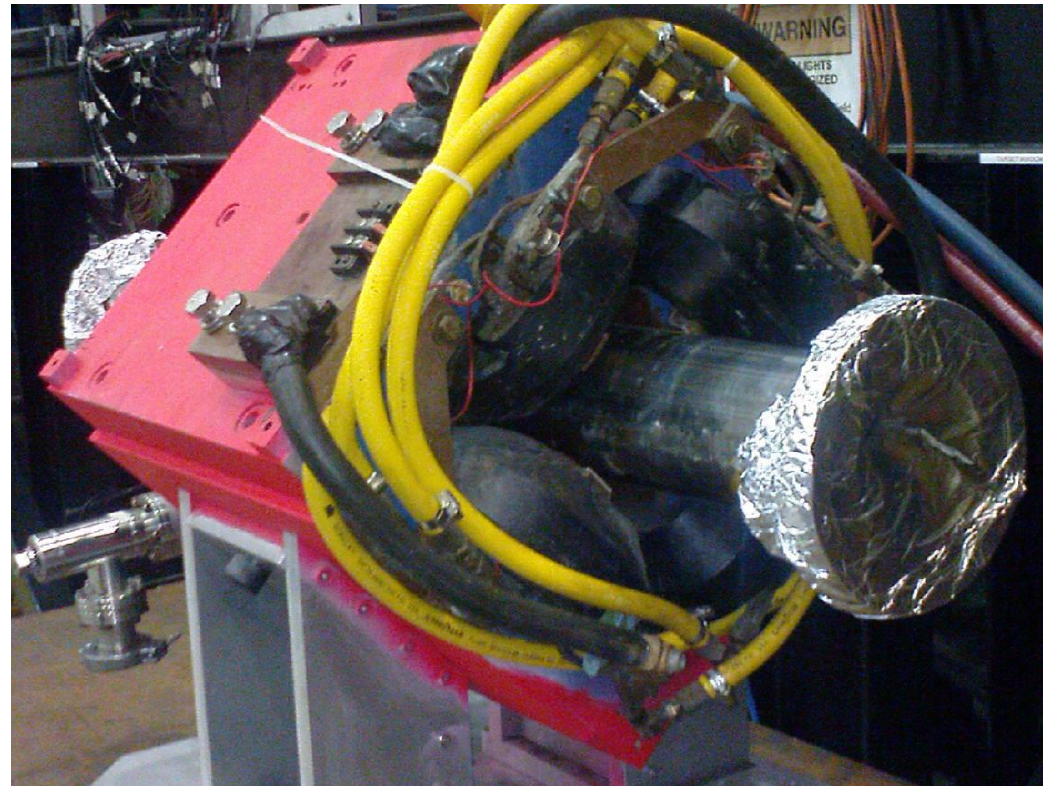
New Quad Q1: TOSCA



Q2=Old Q1=PATSY=MQM1H02 Field Mapping Results

Was done:

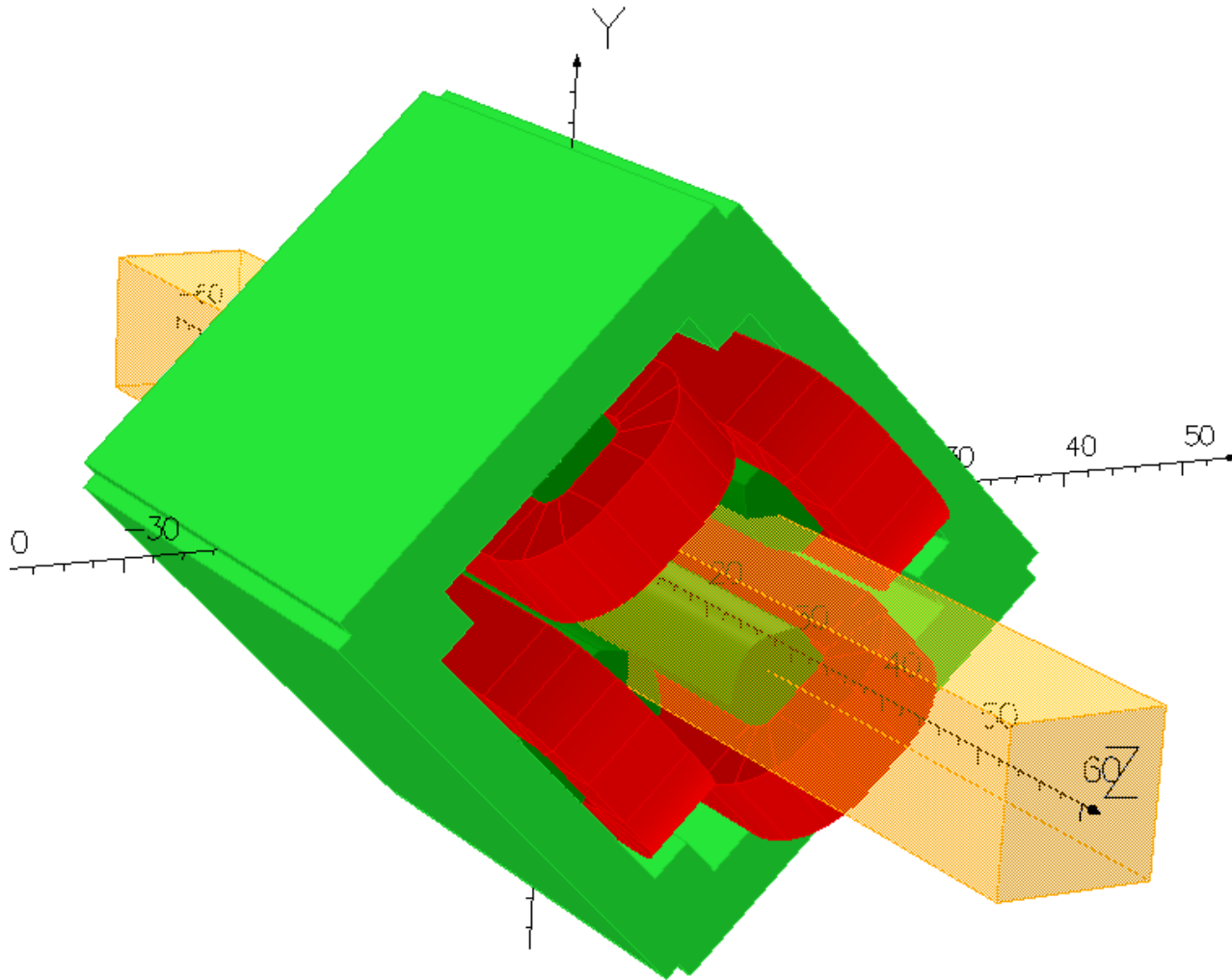
- GL measurement (string R=4.85cm)
- B measurement (Hall probe R~4.85cm) along the quad with fringe fields
- B vs. I measurement near the pole (Hall probe)



Q2=Old Q1=PATSY=MQM1H02 Field Mapping Results

Set Sur.(A)	Readback Current(A)	GL, G measur	Field, G R=4.85cm	Field, G R=5.08cm	HPr G
-300.	-298.92	-52279.4	-5657.64	-5938.18	-5560
-250.	-249.36	-43945.1	-4755.71	-4991.53	-4669
-200.	-198.01	-35298.7	-3820.00	-4009.42	-3750
-150.	-149.68	-26480.6	-2865.71	-3007.81	-2811
-100.	-100.11	-17643.0	-1909.31	-2003.99	-1873
-50.	-50.33	-8755.3	-947.49	-994.48	-930
-10.	-10.49	-1648.3	-178.38	-187.22	-175
0.	0.0	235.7	25.51	26.78	24
10.	10.47	2119.8	229.40	240.78	221
50.	50.25	9237.0	999.62	1049.19	979
100.	100.02	18137.9	1962.87	2060.20	1929
150.	149.70	26986.4	2920.45	3065.26	2877
200.	199.43	35810.7	3875.41	4067.58	3828
250.	249.05	44466.6	4812.15	5050.76	4760
300.	298.92	52321.4	5667.07	5942.95	5604

Quad Q2



Q2=PATSY=MQM1H02

From Ken Baggett (2012)

$I(A)$	$R(cm)$	$BL(kG*cm)$	$B(kG)$	$L_{eff}(cm)$
298.99	4.85	253.5147	5.6684	44.76

Maximal Field at pole:

$$B = 5.6684 / 4.85 * 5.08 = 5.937 kG$$

TOSCA:

$R = 4.85 cm$ $B = 5.659 kG$

Maximal Field at pole: $B = 6.145 kG$

TOSCA grid file: q2_6.145kG.table

X: $-5 cm \div +5 cm$ step $0.25 cm$

Y: $-5 cm \div +5 cm$ step $0.25 cm$

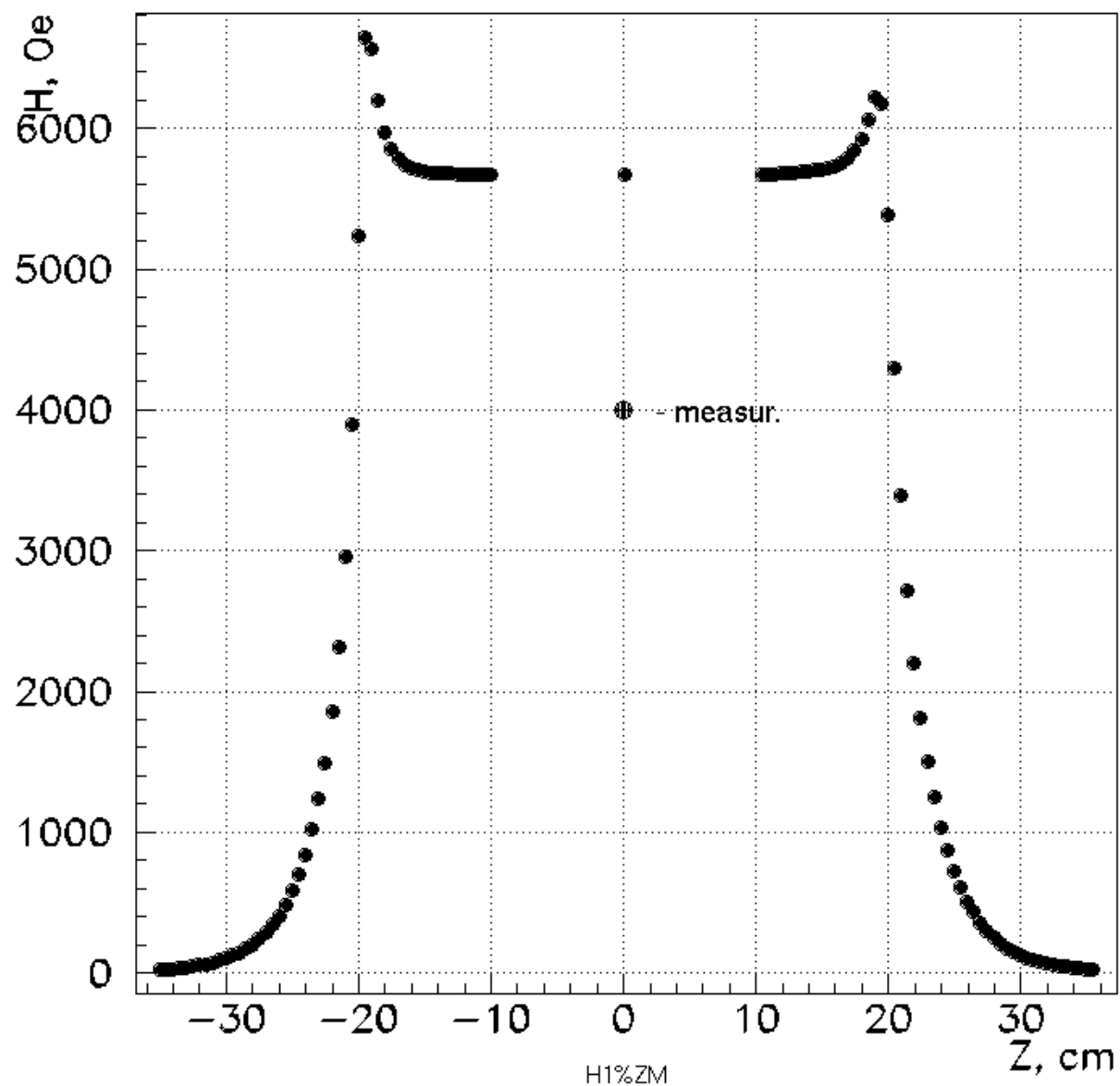
Z: $-60 cm \div +60 cm$ step $0.5 cm$

File size 57MB

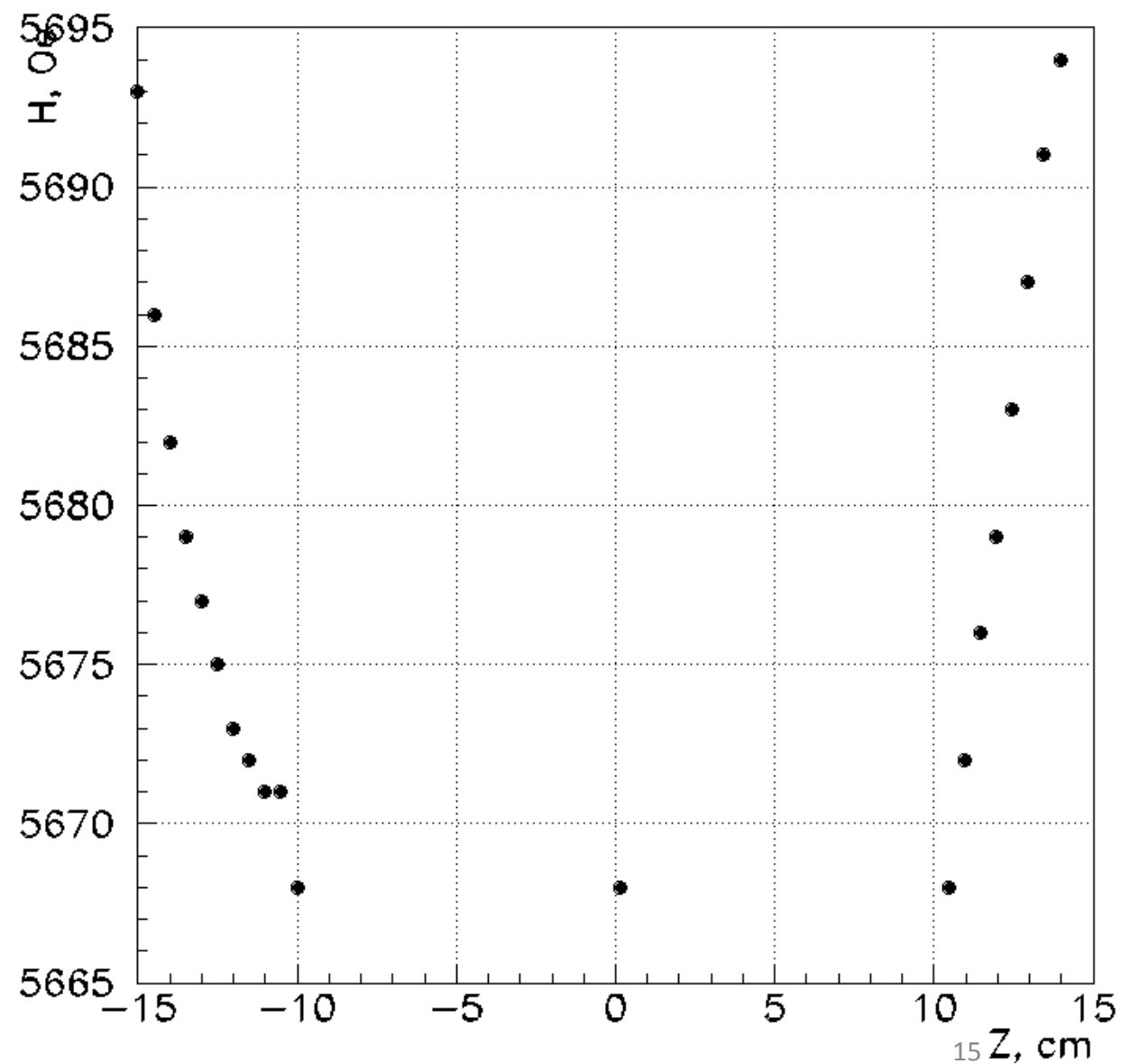
Location: /u/scratch/glamazdi/

Quad Q2: measurement

Q2 Measurement vs. TOSCA

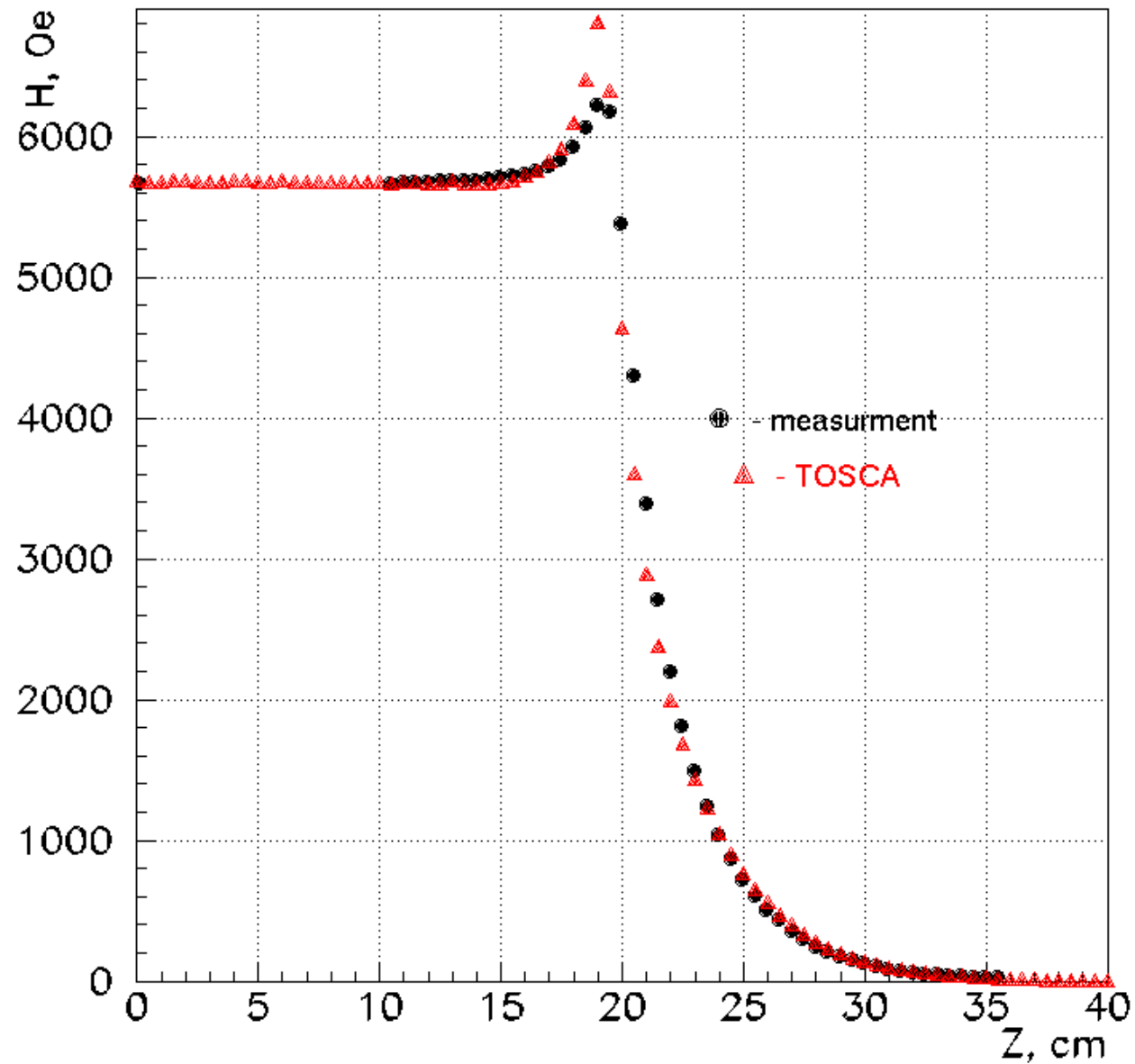


Q2 Measurement vs. TOSCA



Quad Q2: TOSCA

Q2 Measurement vs. TOSCA



8 7 6 5 4 3 2 1

1-50

TOP VIEW 1-50

BEAM DIRECTION →

63.625 REF (SHIELDING INSERTION)

70.25 REF

66.75 STL

9.64 REF

2.36 REF SHLD (60 mm)

4.55 REF (VAC BOX)

11.94

59.75 REF (SHIELDING) (1 in. STL)

76.00 (VACUUM BOX)

11.94

10 EXIST

7

DETAIL 1-4A

11 EXIST

6 EXIST

1 EXIST

64.62 STL

10.83 REF

16.87 REF

9.33 REF

11.97 REF

7.19 REF

9.18 REF

LS VIEW 1-8D (LNG DOWNSTREAM)

RS VIEW 1-3D (LNG UPSTREAM)

ISO VIEW 1-7B

11.81 REF SHLD (300 mm)

12.10 REF (310 mm)

2.36 (60 mm)

#1,360

7 REF

DETAIL VIEW 1-4A

SCALE 1:2 (1-50)

NOTES:

- ITEM 7 - THE SHIELDING INSERTION ASSY IS A NEW ASSEMBLY. ALL OF THE OTHER ITEMS ARE EXISTING EQUIPMENT SHOWN FOR REFERENCE.
- THE SHIELDING INSERTION (ITEM 7) IS TO BE COVERED ALONG 2 (BEAM DIRECTION) WITHIN THE BOPE OF THE SHIELDING (ITEM 2) OF THE VACUUM BOX (ITEM 3).

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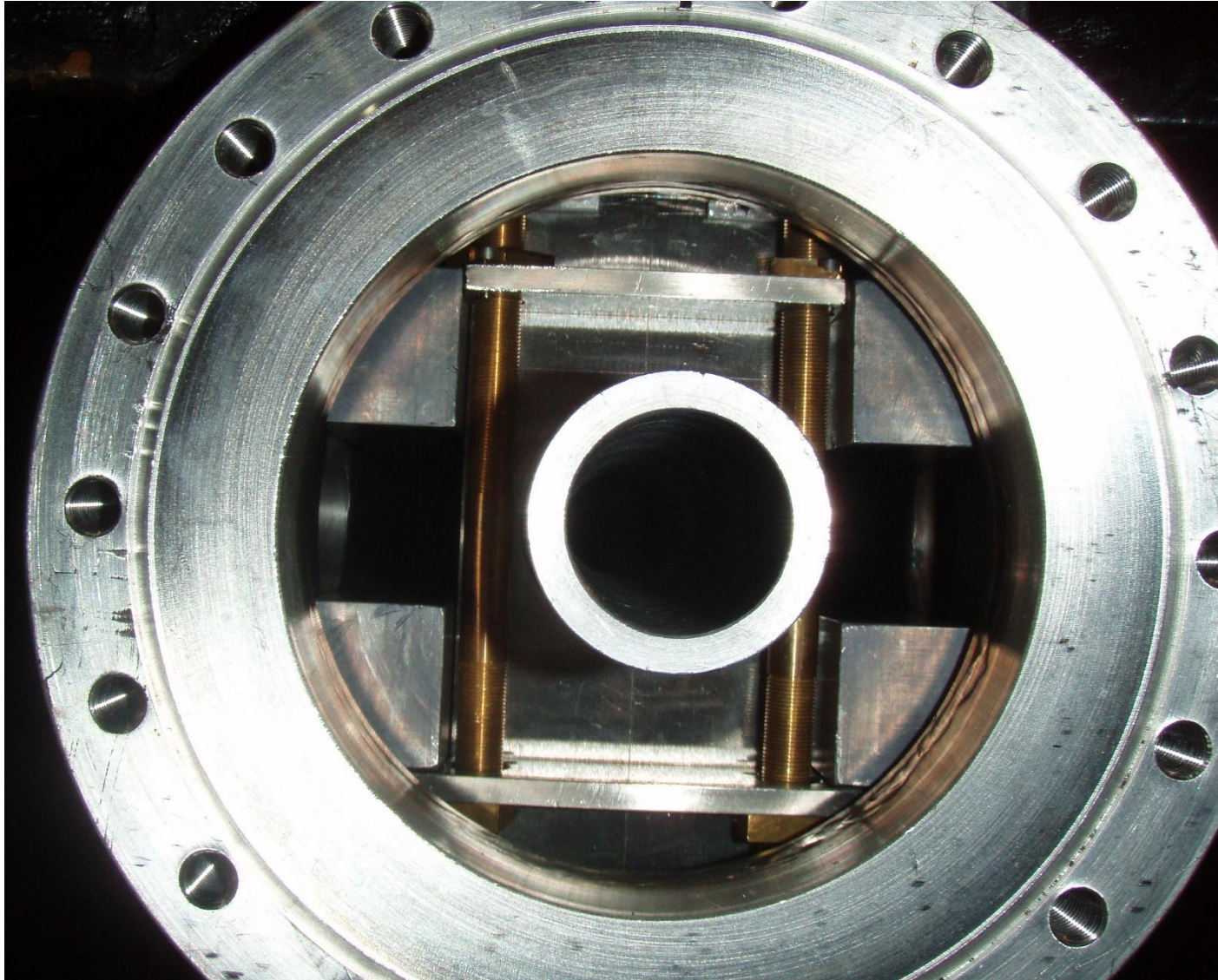
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NO.	DATE	BY	CHKD	DESCRIPTION
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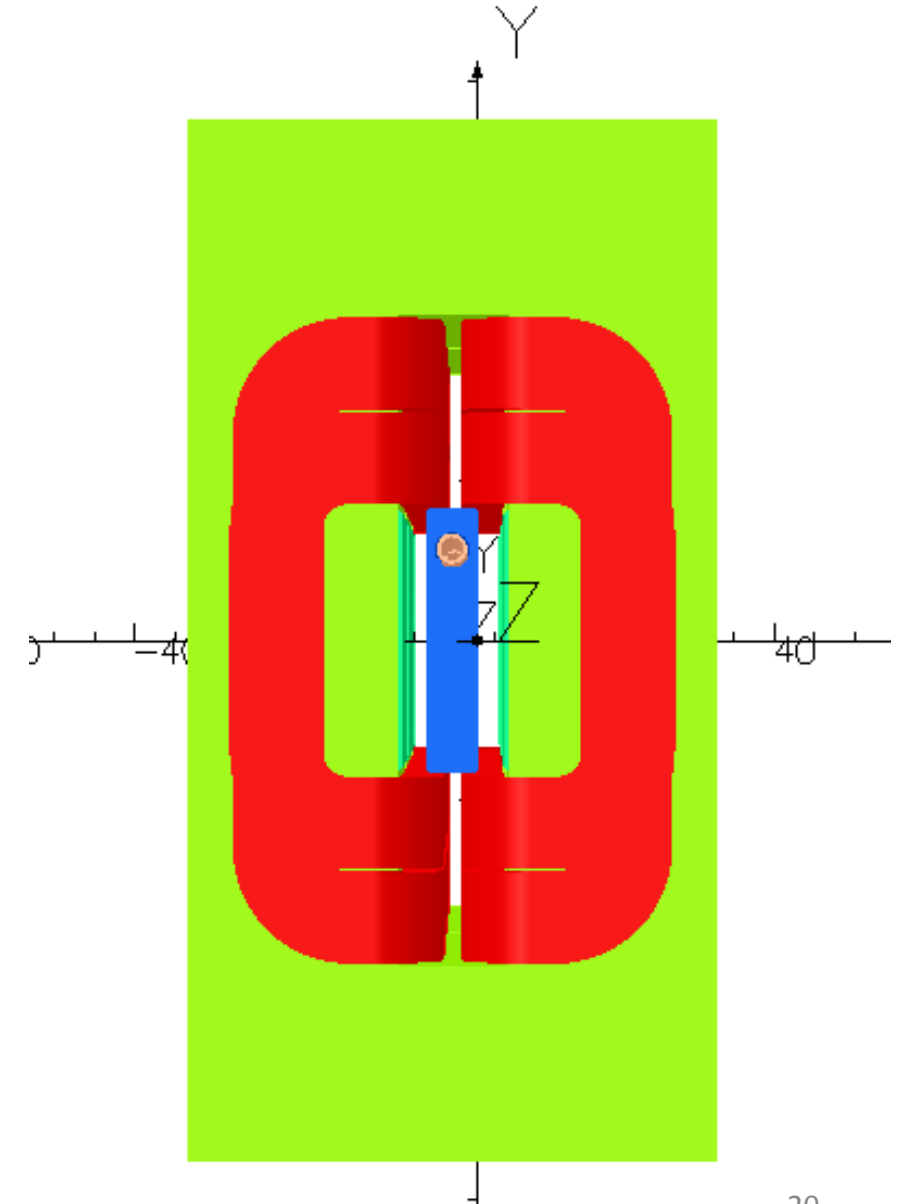
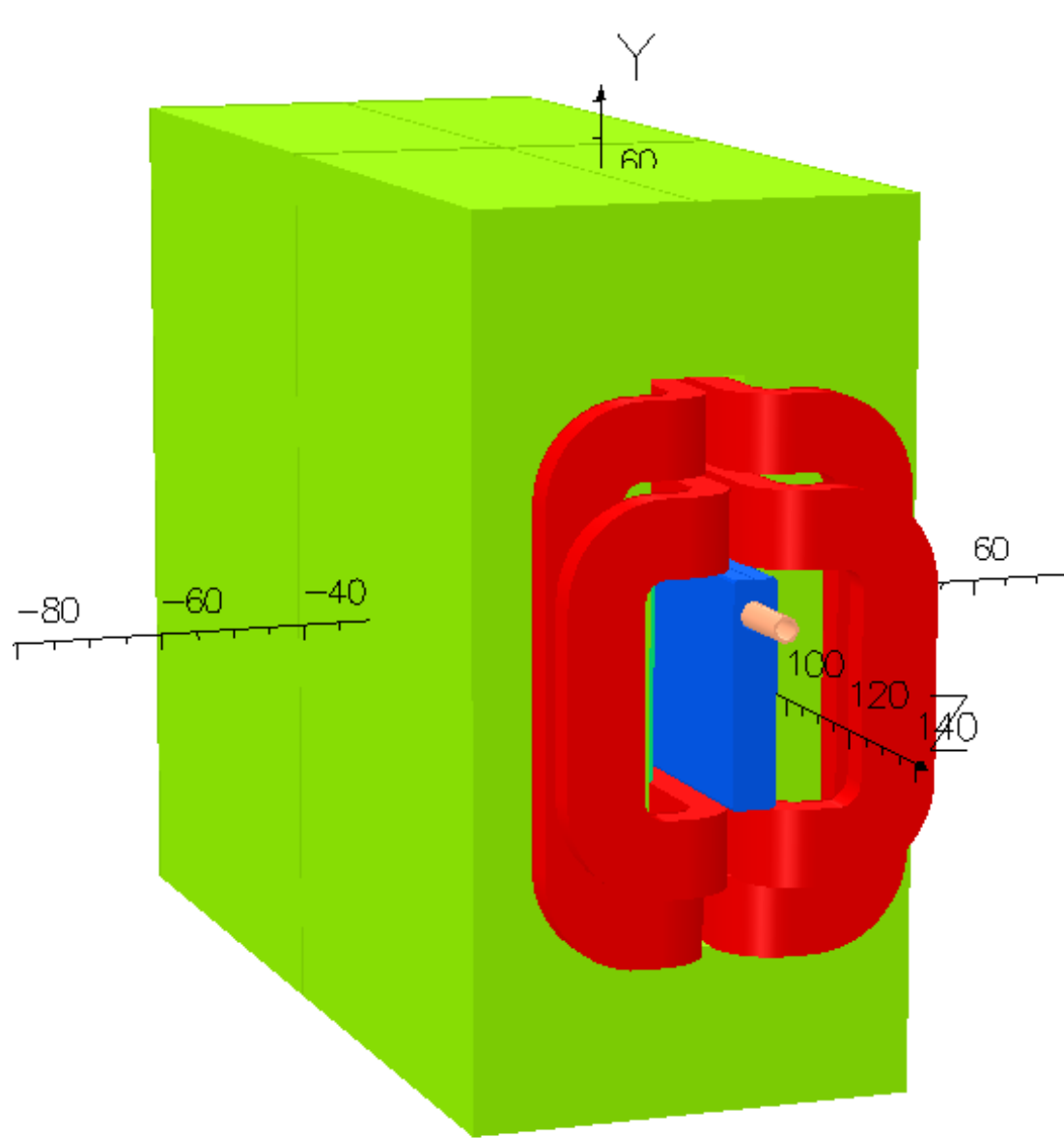
Moller Dipole LILLY Upgrade



Moller Dipole Upstream Collimator

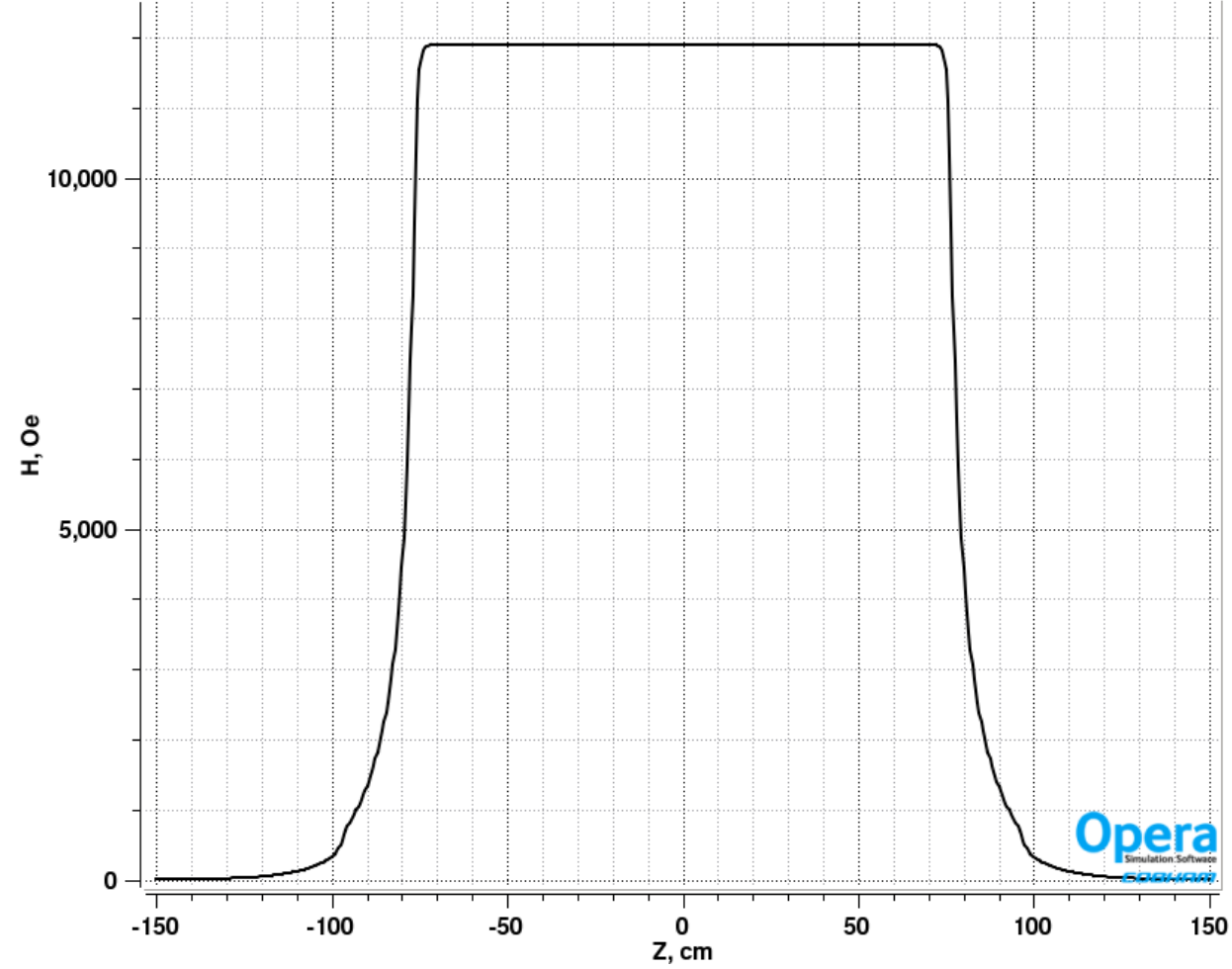


Moller Dipole LILLY: TOSCA

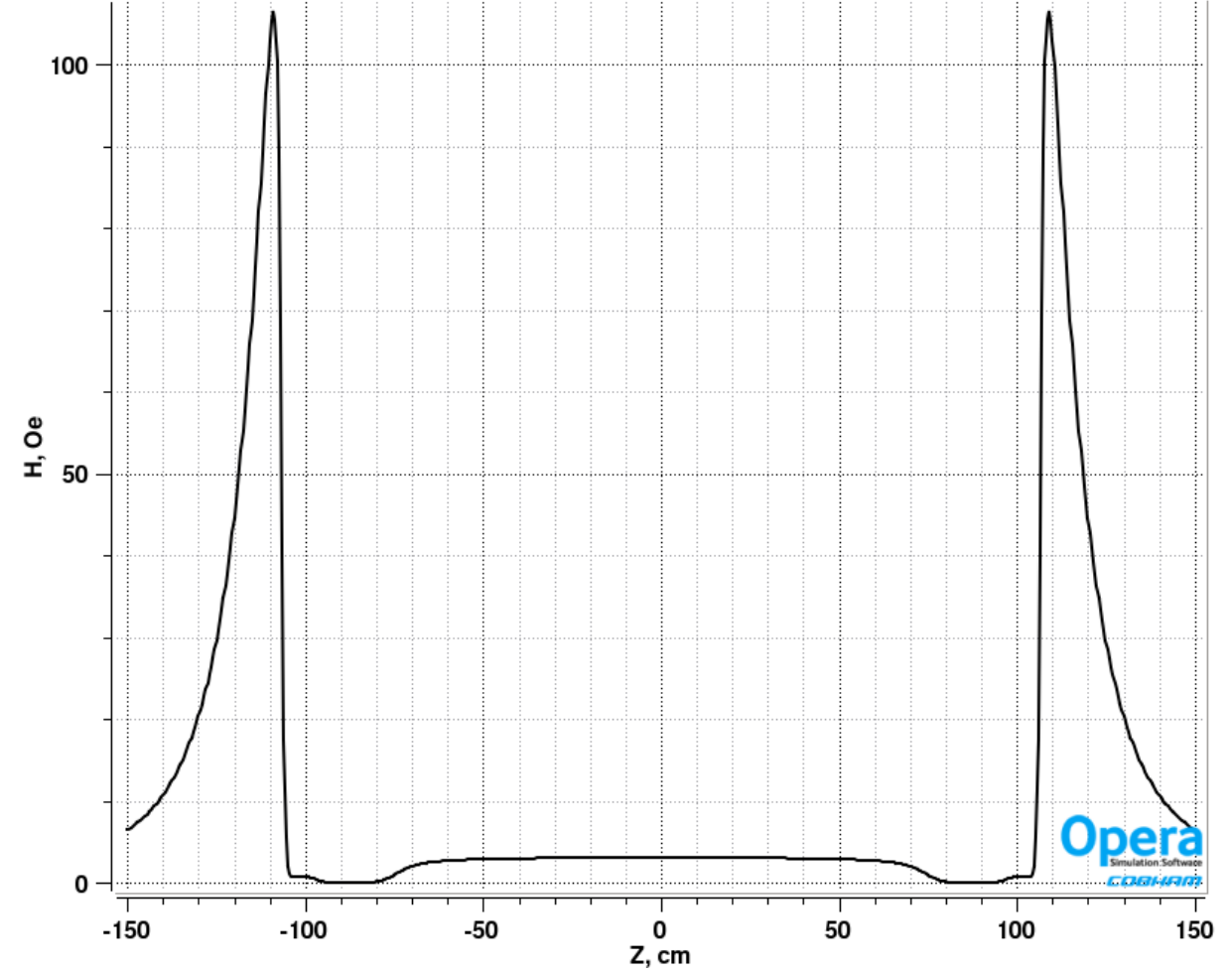


LILLY: TOSCA Result

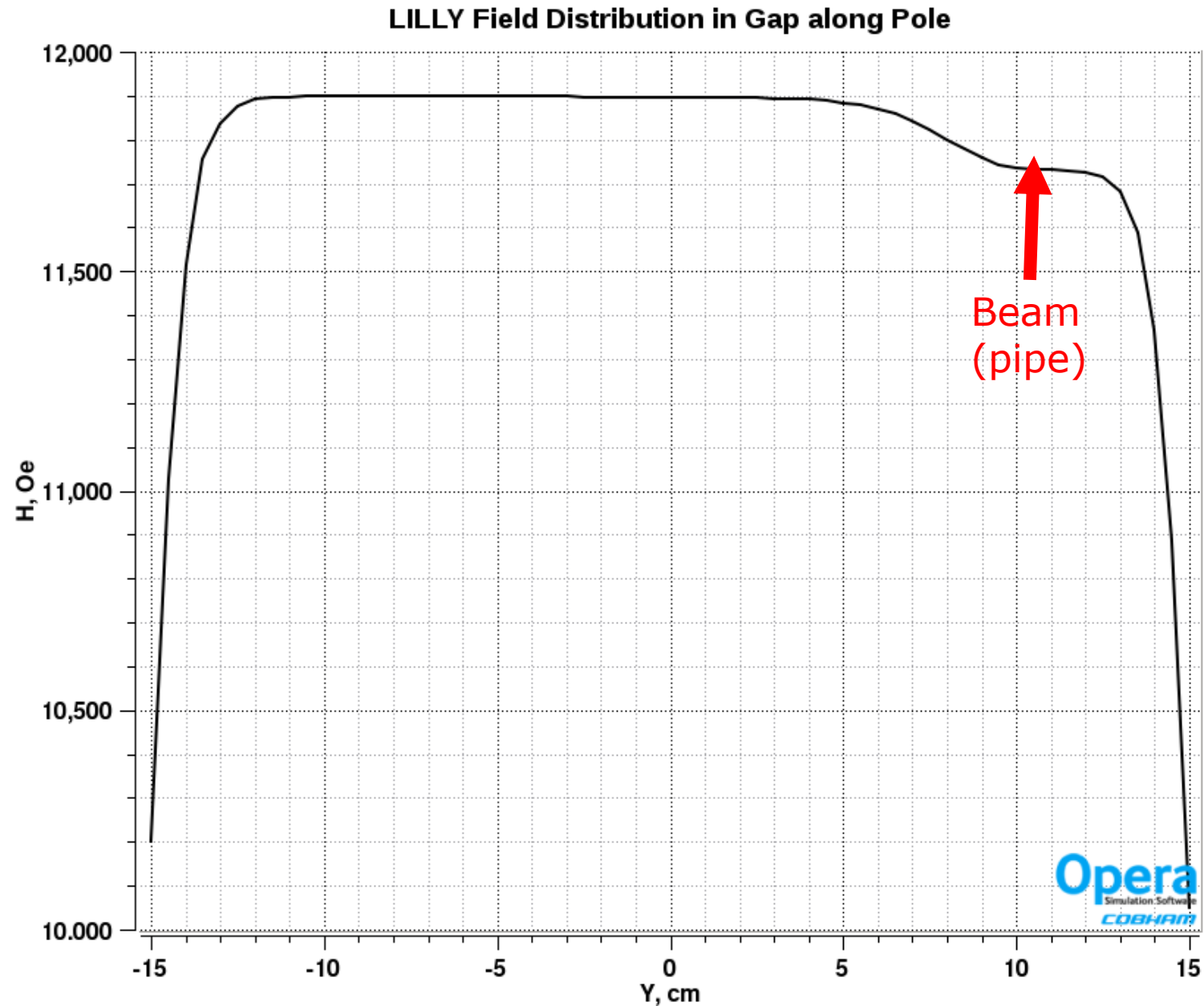
LILLY Field Distribution in Gap along Z



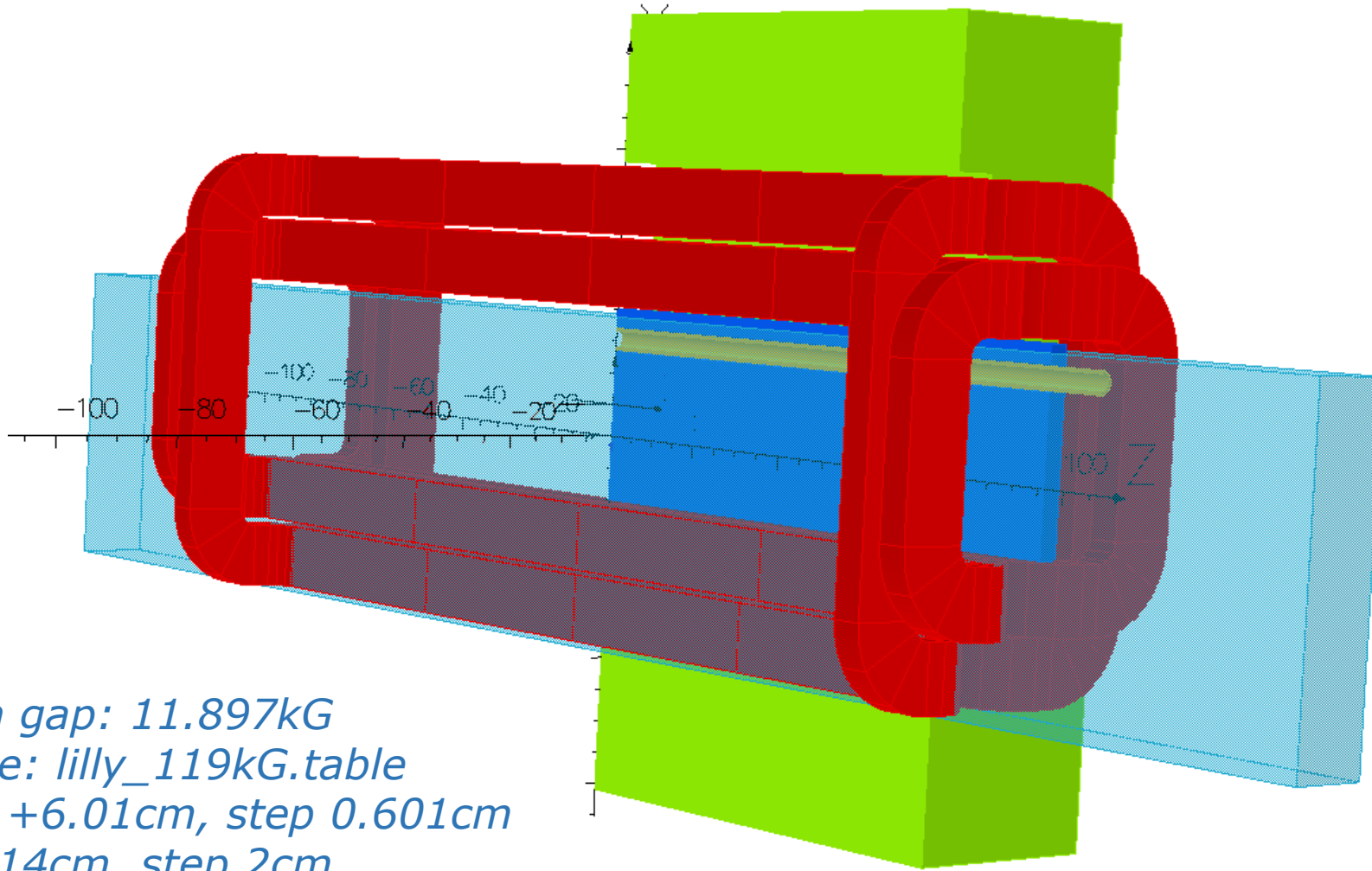
LILLY Field Distribution in Pipe



LILLY: TOSCA Result



Moller Dipole LILLY: TOSCA Grid



TOSCA field in gap: 11.897kG
TOSCA grid file: lilly_119kG.table
X: -6.01cm ÷ +6.01cm, step 0.601cm
Y: -38cm ÷ +14cm, step 2cm
Z: -150cm ÷ +150cm, step 2cm
File size 12MB, Location: /u/scratch/glamazdi